

Endocrine Histology Laboratory Manual

Touro University
B. Puder, Ph.D.

Reading Assignment: Color Textbook of Histology, Garnter & Hiatt,
Chapter 13, pp. 303 – 325, Chapter 18, pp. 420 – 422.

Objectives

Upon completion of the endocrine laboratory, you should be able to answer the following objectives:

Pituitary gland (hypophysis)

1. Identify the pituitary gland on a digital image and identify the following structures:
 - A. Pars distalis (anterior lobe)
 - B. Pars intermedia
 - C. Pars nervosa (posterior lobe)
 - D. Infundibular stem
 - E. Pars tuberalis
 - F. Rathke's cysts
2. Identify the following cell types in the pars distalis (anterior lobe):
 - A. Acidophils
 - B. Basophils
 - C. Chromophobes
3. Describe the sites of production and the mechanisms involved in the regulation of synthesis and release of the following hormones:
 - A. Oxytocin
 - B. Vasopressin (antidiuretic hormone)
 - C. Growth hormone (somatotropin)
 - D. Prolactin (lactogenic hormone, mammotropin)
 - E. Thyroid stimulating hormone (thyrotropin)
 - F. Follicle stimulating hormone
 - G. Luteinizing hormone
 - H. Adrenocorticotrophic hormone
 - I. Melanocyte stimulating hormone
4. Describe how releasing factors or inhibitory factors from the hypothalamus control hormone secretion from the hypophysis.
5. Identify the pineal gland on a digital image and identify the following structures:
 - A. Pinealocytes/glial cells
 - B. Calcium rich (calcareous) concretions called "brain sand", psammoma bodies, or corpora arenacea
6. State the hormone that is produced by the pinealocytes.
7. Describe the significance of the brain sand in the pineal gland.

8. Identify the thyroid gland on a digital image and identify the following structures:
 - A. Thyroid follicles consisting of:
 - i. colloid of thyroglobulin
 - ii. follicle cells enclosing colloid
 - B. Parafollicular cells or C cells
9. State the hormone which stimulates the thyroid follicular cells and list the hormones that are secreted from the thyroid follicular cells and describe those hormones' function.
10. State the hormone that is secreted from the parafollicular cells and describe their function.
11. Locate the parathyroid on a digital image and identify the following structures:
 - A. Chief cells
 - B. Oxyphil cells
12. Briefly describe the function of the chief cells. What hormone is produced by the chief cells? What is that hormone's function?
13. Identify the adrenal gland on a digital image and identify the following structures:
 - A. Adrenal cortex
 - B. Adrenal medulla
 - C. Connective tissue capsule
 - D. Zona glomerulosa
 - E. Zona fasciculata
 - F. Zona reticularis
14. List the hormones that are secreted from the zona glomerulosa, zona fasciculata, and zona reticularis
15. Identify the endocrine glands (islets of Langerhans) within the pancreas on a digital image.
16. State the 5 different types of cells within the endocrine pancreas.
(You will not be able to identify these 5 cell types on an H&E stained section.)
State the hormones that are secreted from the 5 cell types and describe their general function.

***** You are responsible for identifying on digital images all bold faced and italicized terms on the following pages of this endocrine lab manual for the lab practical.**

The Endocrine System

General Information

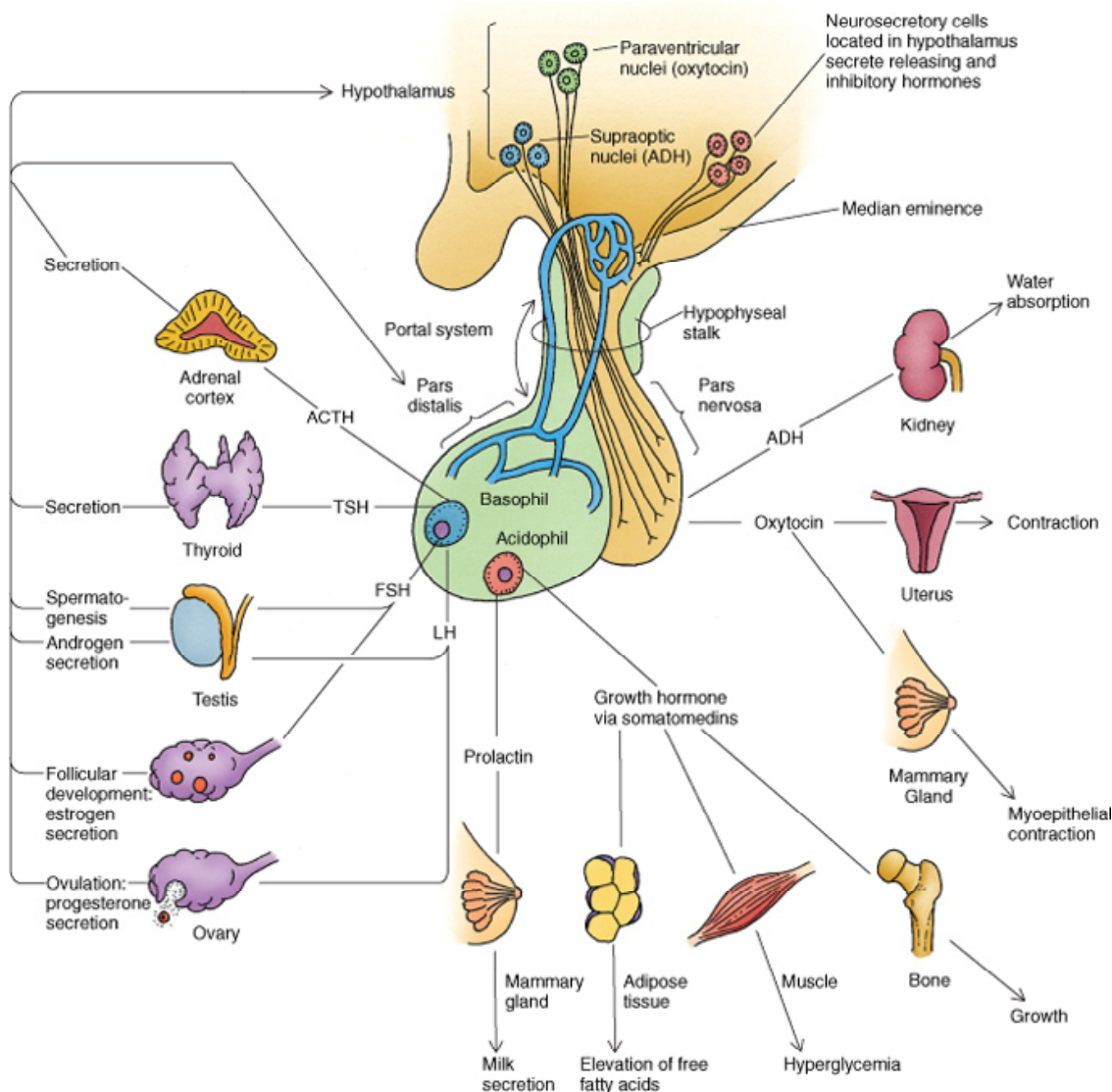
The Endocrine system consists of a group of glands, organs, and cells that secrete chemical messengers (hormones) into the bloodstream to elicit various effects.

The endocrine system helps to regulate various metabolic activities in organs and tissues to maintain homeostasis.

The endocrine system is a slow and diffuse system (as compared to the nervous system which can send AP signals at rates of 200 mph.).

The pituitary gland (hypophysis) is a key component of the endocrine system. The posterior pituitary (pars nervosa) receives neural input via axons from the hypothalamus. The terminal endings of these axons release the hormones oxytocin and vasopressin (antidiuretic hormone) into the posterior pituitary. These 2 hormones are then released into the vasculature to elicit their effects. Other hormones called releasing hormones are produced in the hypothalamus and sent via the vasculature to the anterior pituitary where the releasing hormones influence the cells of the anterior pituitary to release various hormones into the vasculature.

The pituitary is influenced by many feedback loops from target organs and cells.



Images 1 - 4

Pituitary gland (hypophysis)

The **pituitary gland** consists of 2 large areas:

1. **Adenohypophysis**

consists of:

- pars distalis (anterior lobe)**
- pars intermedia (intermediate lobe)**
- pars tuberalis**

2. **Neuropophysis**

consists of:

- pars nervosa (posterior lobe)**
- infundibular stem**

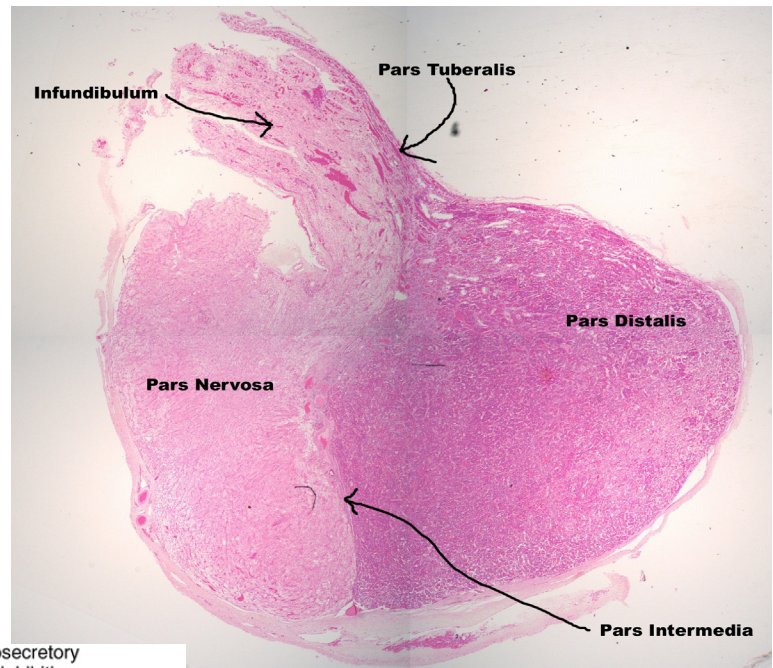


Image 1: 2X image of pituitary gland

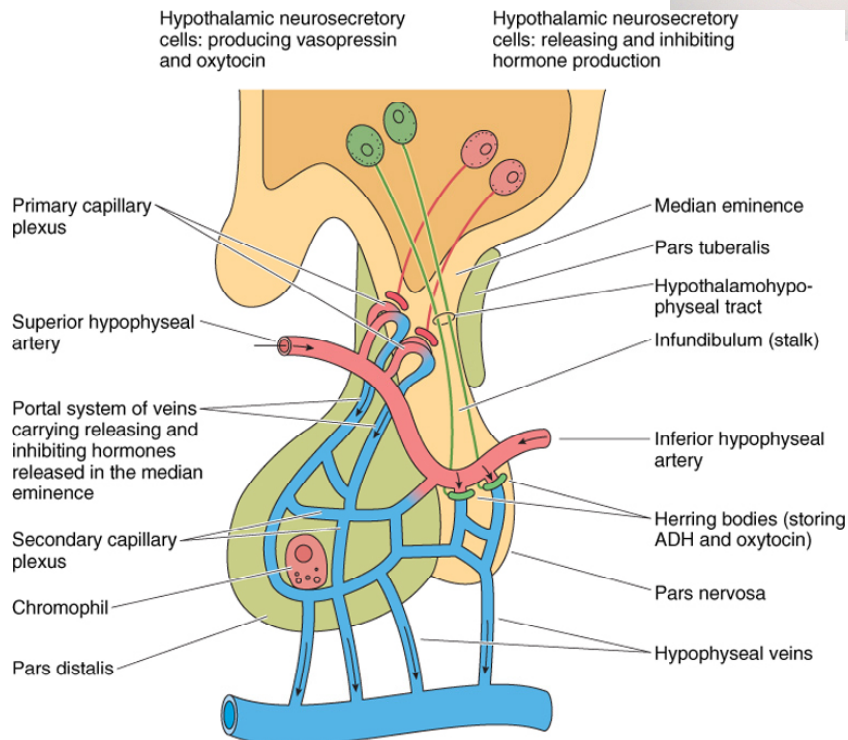


Diagram from Gartner & Hiatt text depicting the vasculature of the pituitary. Note that this image is orientated in the opposite direction as Image 1.

On the higher magnification image, look in the **anterior pituitary (pars distalis)** for:

Acidophils (pink granules)

Basophils (purple granules)

Chromophobes (pale cytoplasm)

What are the names of the hormones produced by the acidophils and basophils?

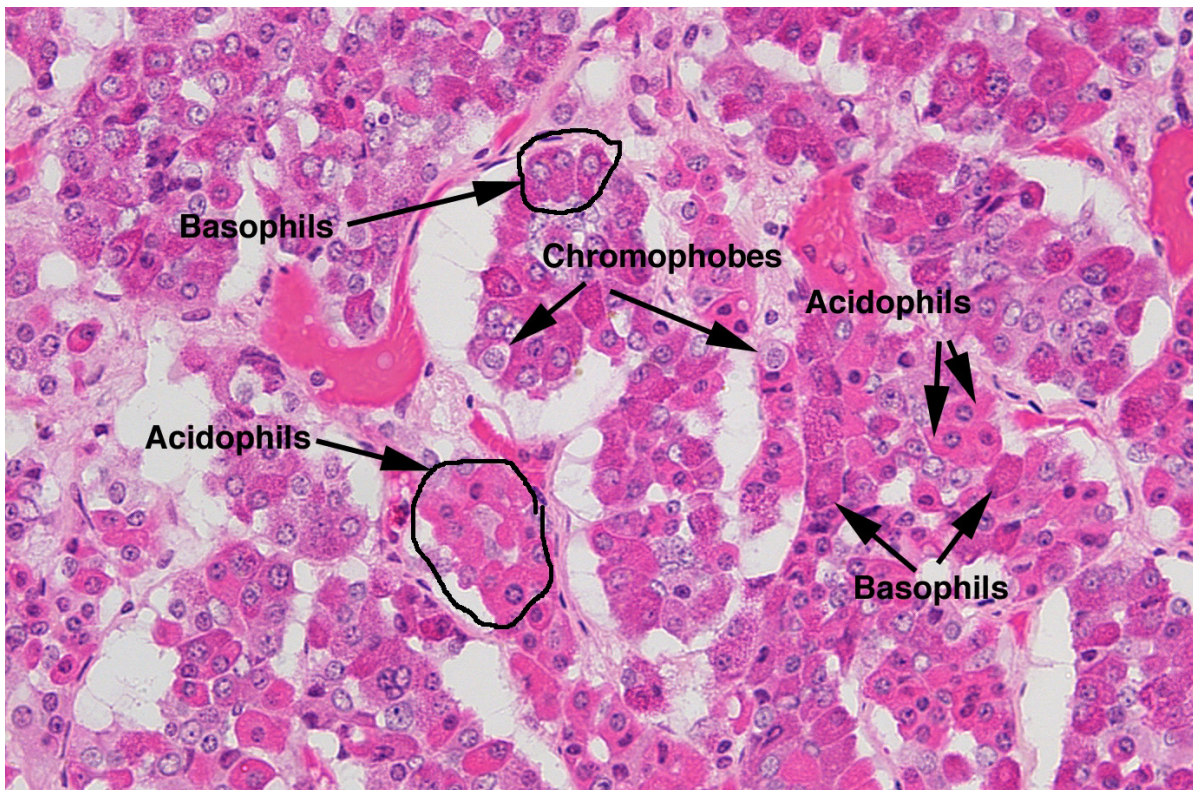


Image 2: 40X image of the anterior pituitary (pars distalis)

Look in the **pars intermedia** for colloid-filled **Rathke's cysts**.

Identification of these cysts are diagnostic for the **pars intermedia**.

Melanocyte-stimulating hormone (MSH) may be produced (as a cleavage product of POMC [pro-opiomelanocortin]) in the intermediate lobe, however, it may be produced in the anterior lobe as well.

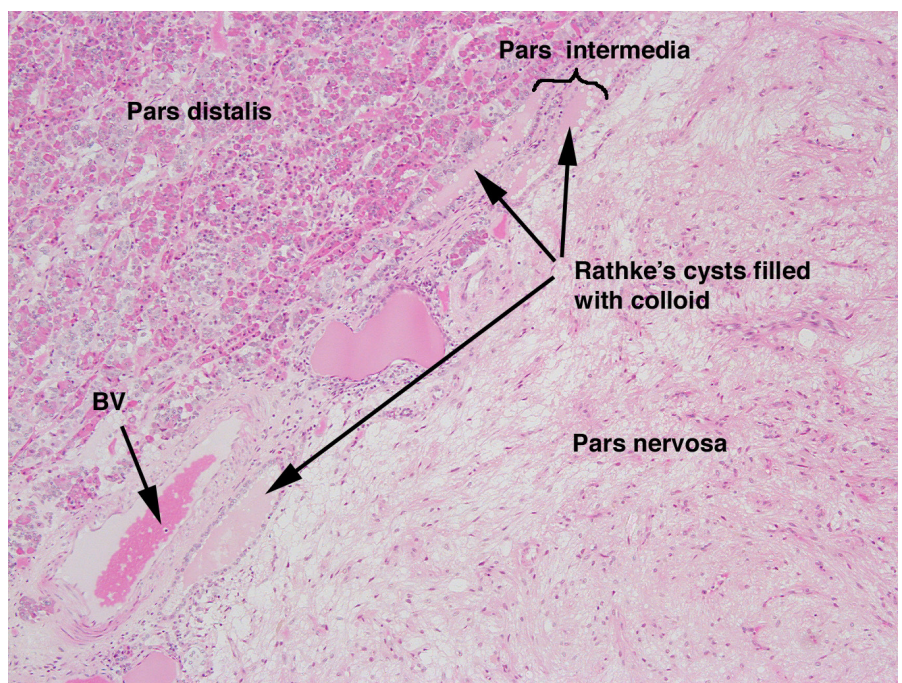


Image 3: 10X image of the intermediate lobe (pars intermedia)

The **posterior pituitary (pars nervosa)** contains glial-like cells called **pituicytes**.

Recall that the paraventricular and supraoptic nuclei of the hypothalamus project their axons into the **posterior pituitary**. The terminal bouton axonal endings are called **Herring bodies** and they release either oxytocin or vasopressin (antidiuretic hormone) from the **Herring bodies** into the vasculature so that oxytocin or vasopressin can travel to target organs to elicit their effect.

Herring bodies are seen as granular, ovoid structures that stain gray to slightly eosinophilic.

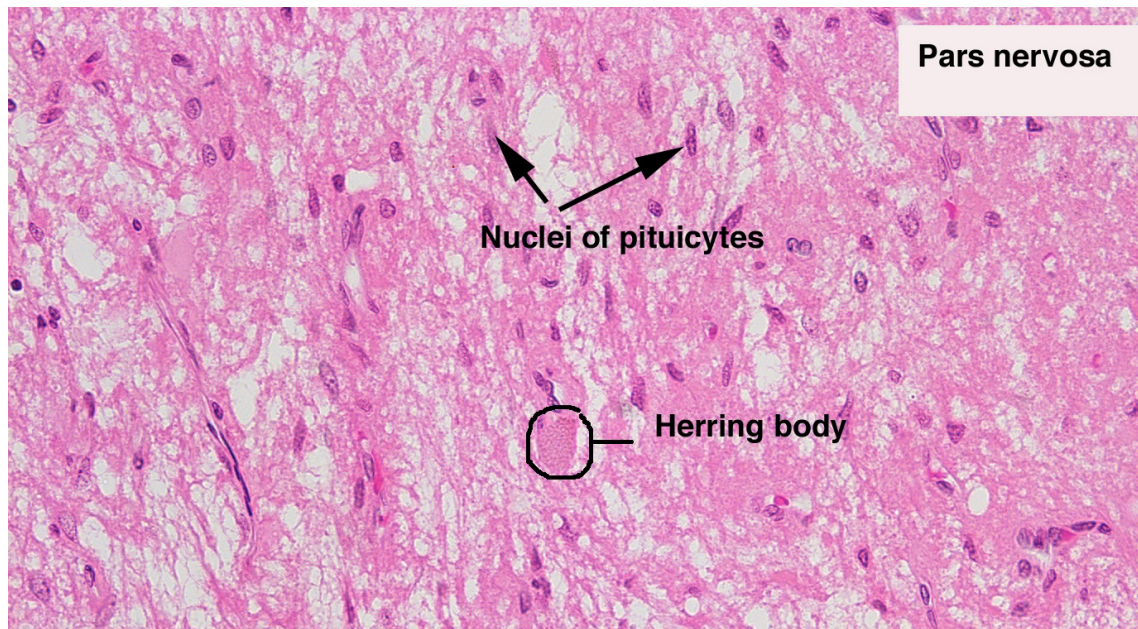


Image 4: 40X image of posterior pituitary (pars nervosa)

Image 5 Pineal gland

The 2 basic cell types in the **pineal gland** are **pinealocytes** and **glial cells**. We cannot distinguish between the 2 cell types with this staining technique.

In the adult pineal gland, you will see calcareous deposits called **corpora arenacea** or "**brain sand**" or **psammoma bodies**. These calcium deposits are located in the connective tissue septae of the gland and increase with age. These calcium deposits also make a great radiologic mid-sagittal brain landmark when imaging the brain.

The pinealocytes synthesize melatonin from serotonin (5 HT) during dark exposure. Melatonin inhibits growth hormone and gonadotropins and also makes individuals feel sleepy.

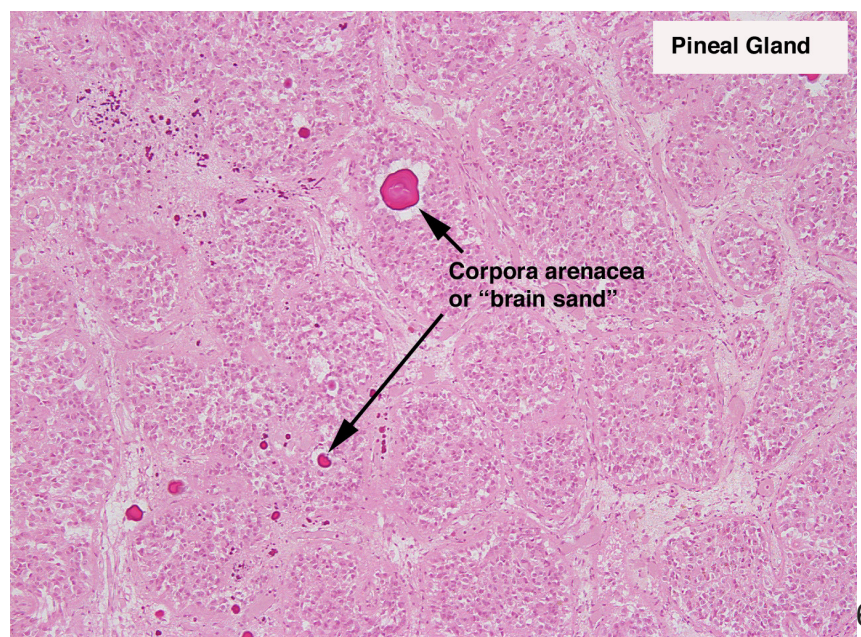


Image 5:
10X image of the pineal gland

Image 6

Thyroid

The **thyroid gland** is located in the anterior aspect of the neck.

Upon viewing the **thyroid**, observe the cuboidal **follicular cells** that surround **colloid (thyroglobulin)** being stored in the follicular lumen. Identify **parafollicular cells** located in the interfollicular spaces.

The **thyroid gland** secretes the hormones triiodothyronine (T_3), thyroxine (T_4), to increase metabolism and calcitonin to store calcium in bones.

T_3 and T_4 are stored in the colloid and are bound to thyroglobulin.

The thyroid gland is influenced by thyroid stimulating hormone (TSH) from basophils of the anterior pituitary.

This image can be confused with the lactating mammary gland (good thing we don't have that slide in this course!!) However, you can tell the 2 slides apart because the thyroid gland contains no ducts (a characteristic of all endocrine organs), very few connective tissue septae and very little fat.

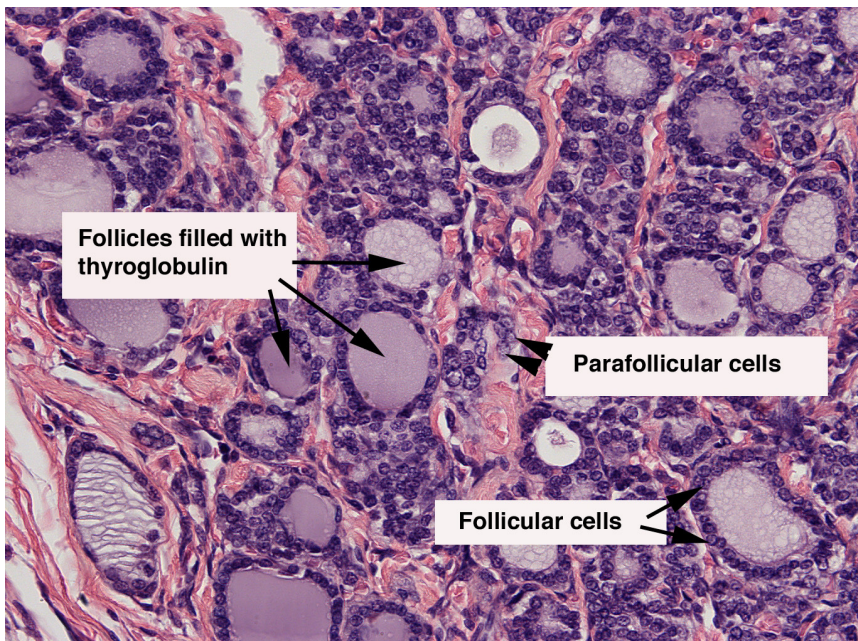
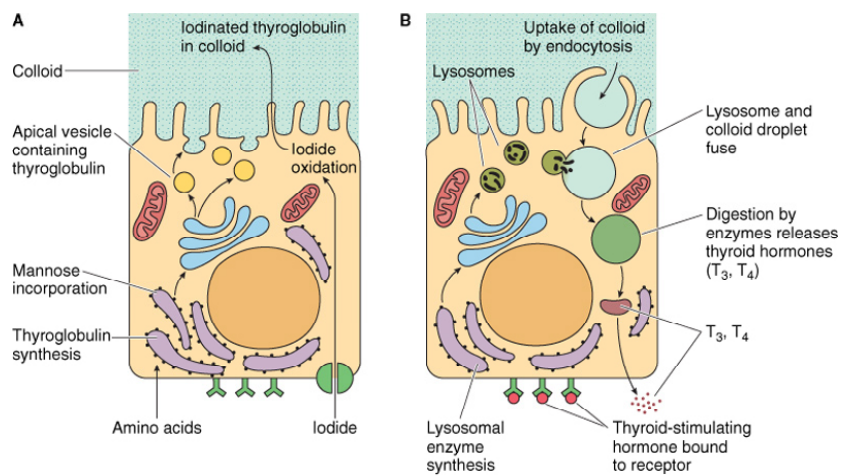
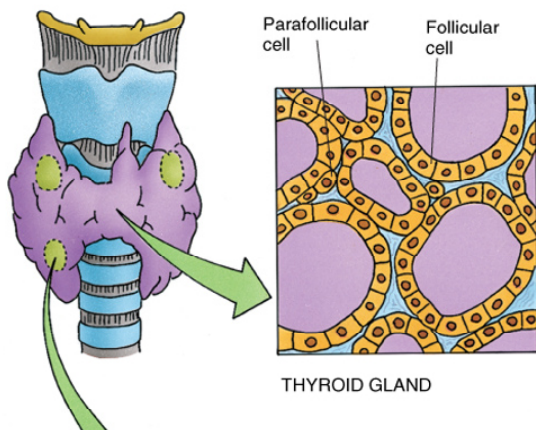


Image 6: 40X image of thyroid gland

Follicular cells produce T_3 and T_4 which is stored bound to thyroglobulin in the follicular lumen.

Parafollicular cells produce calcitonin.



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Diagrams from Gartner & Hiatt text depicting thyroid gland (left) and follicular cells that produce, store and release T_3 and T_4 (right image).

Image 7

Parathyroid glands

These pea-sized glands are located on the posterior aspect of the thyroid, one on each of the upper and lower right and left sides. The glands are surrounded by a capsule that continues as septae that divides the organ into lobules. The **parathyroid cells** are in cords or sheets separated by capillaries and connective tissue.

You can distinguish between 2 cell types in the parathyroid glands:

1. **Chief (principal cells)**

- grouped together, small and have a central spherical nucleus
- produce parathyroid hormone (PTH) to elevate calcium serum levels

2. **Oxyphil cells**

- large eosinophilic cells, appear in clusters
- thought to be precursors to chief cells
- Oxyphil and adipose cells increase with age

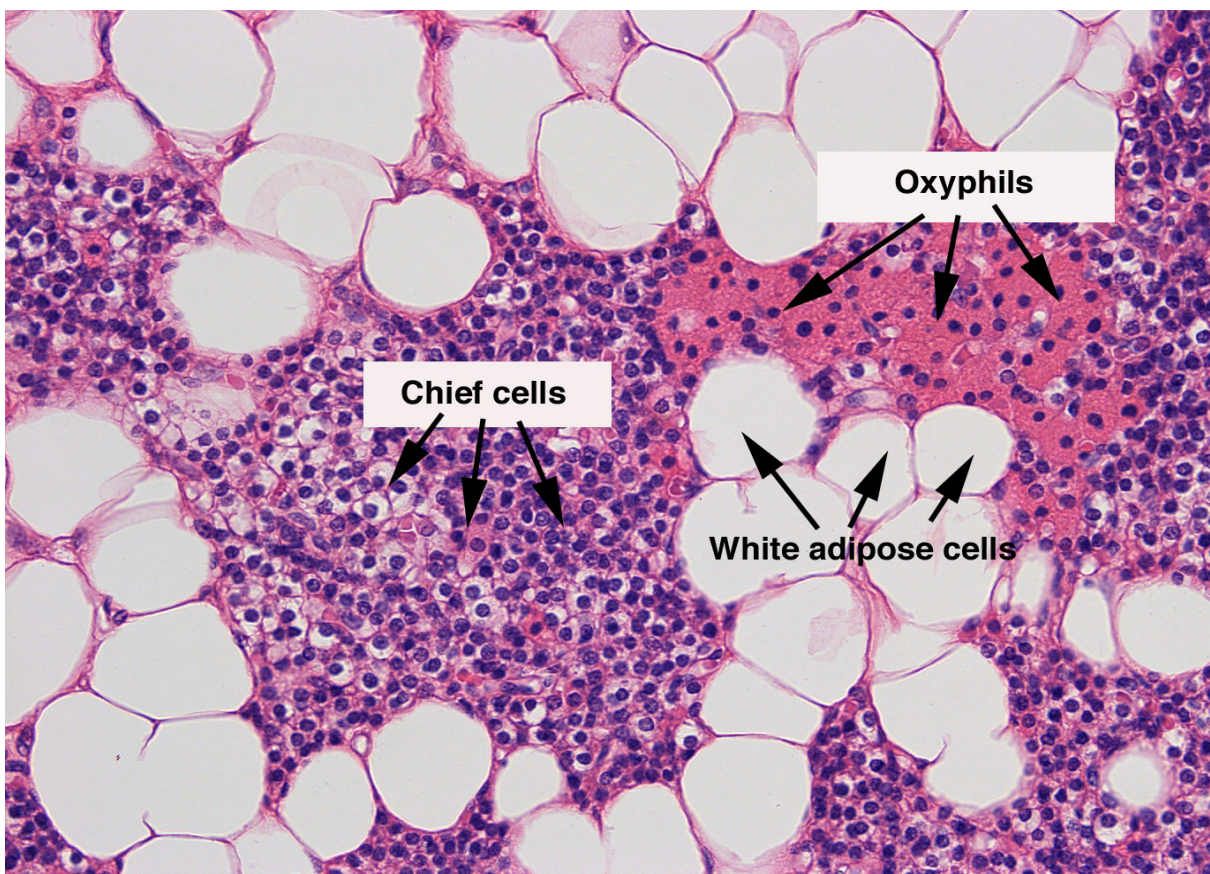


Image 7: 40X image of parathyroid gland

Images 8 - 10

Adrenal (suprarenal) glands

These glands are located on the superior pole of the kidneys.

They consist of a **cortex** derived from mesoderm and a **medulla** derived from neural crest cells.

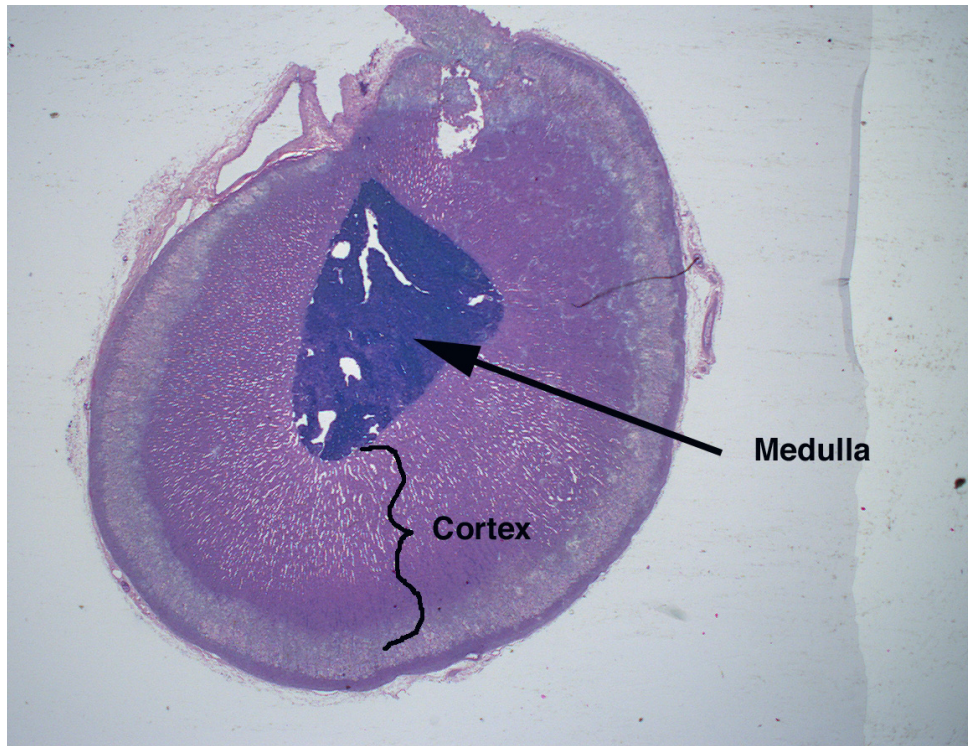


Image 8: 2X image of adrenal gland - chromic stain

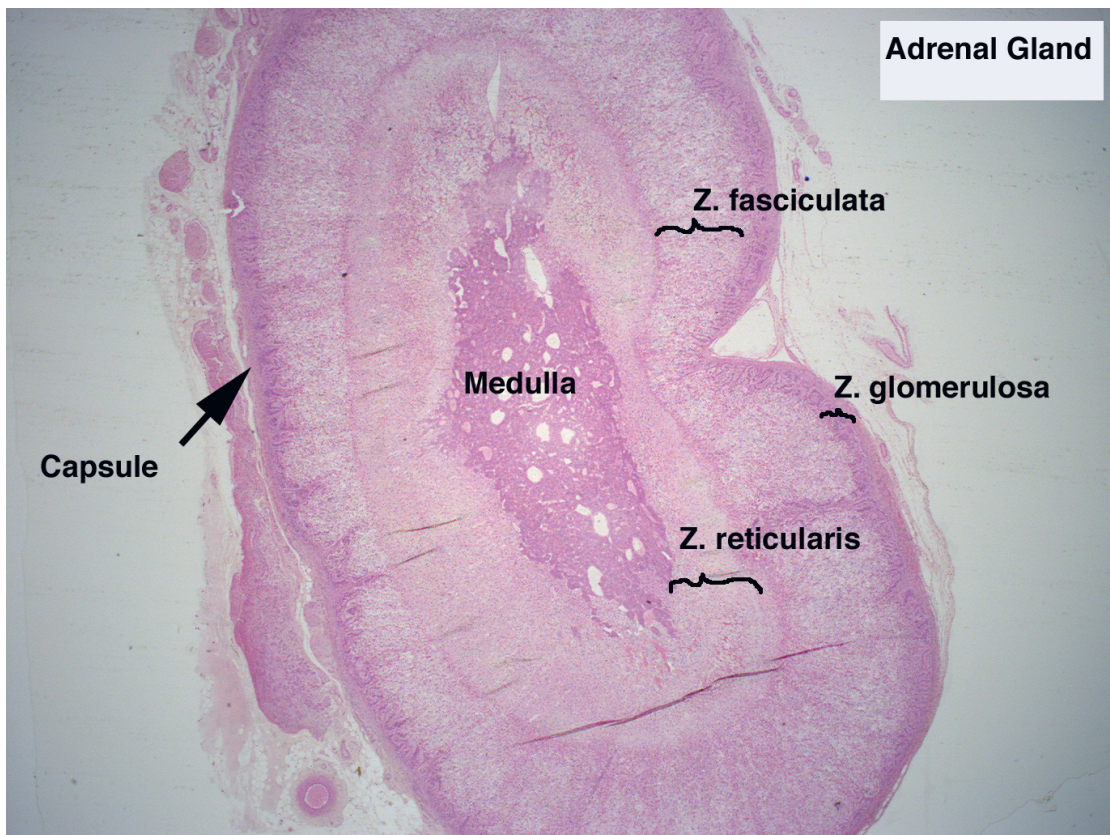


Image 9: 2X image of the adrenal gland - H&E stain

The **adrenal cortex**

There are 3 zones to the adrenal cortex:

1. **Zona glomerulosa**

Outermost region, consisting of flattened cells arranged in loops and hairpin turns
Secretes aldosterone (a mineralcorticoid) - for electrolyte homeostasis and osmotic balance of urine

Stimulated by ACTH and angiotensin II

2. **Zona fasciculata**

Middle region of the adrenal cortex, cells are arranged in straight cords

Secretes: a. cortisol - for carbohydrate, protein and fat metabolism

b. weak androgens

Stimulated by ACTH

3. **Zona reticularis**

Innermost region of adrenal cortex, cells are arranged in anastomosing cords

Secretes cortisol and weak androgens

Stimulated by ACTH

The **medulla** (not visible in image below)

Because it is derived from neural crest cells, it can be considered a modified sympathetic ganglion. Cells in the medulla secrete epinephrine and some norepinephrine.

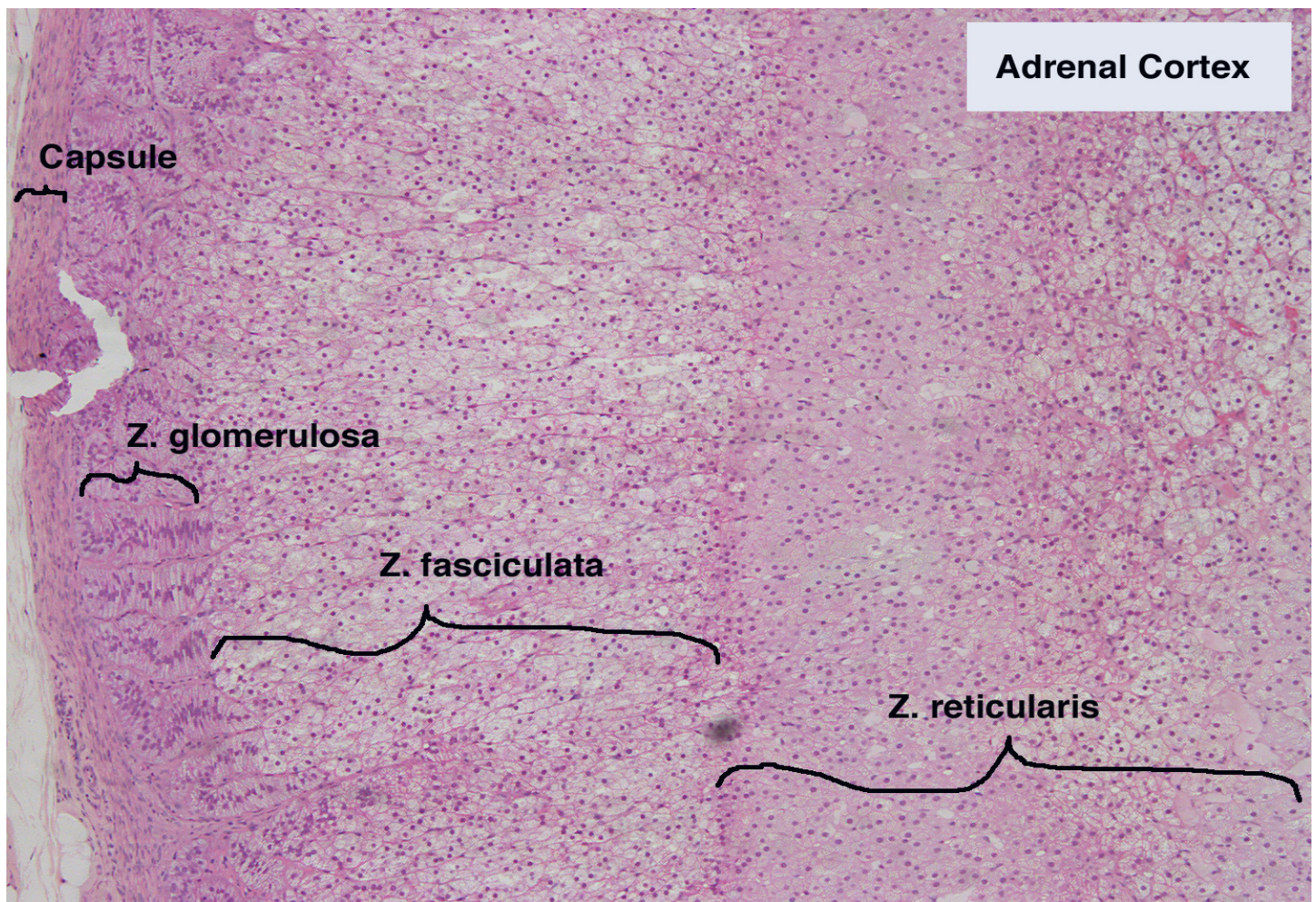


Image 10: 10X image of the 3 zones of the adrenal cortex

Images 11 -13

Endocrine pancreas

The pancreas consists of a endocrine portion and an exocrine portion. Today we will only be identifying the endocrine pancreas which is part of the endocrine system.

(You will identify parts of the exocrine pancreas later when you study the digestive system.)

Within the exocrine pancreas are clumps of cells which have an endocrine function called **pancreatic islets or islets of Langerhans**.

The **pancreatic islet cells** appear less granular and have greater vascularity than the exocrine pancreas.

There are 5 different cell types within the endocrine pancreas, however, we cannot distinguish them using this H&E stain. Those 5 cell types are named: alpha cells, beta cells, delta cells, and G cells, and PP cells.

<u>Cell type</u>	<u>Secretion</u>	<u>Function</u>
alpha	glucagon	increase blood glucose levels
beta	insulin	decrease blood glucose levels
delta	somatostatin and vasoactive intestinal peptide(VIP)	inhibits alpha and beta cells/decrease motility in GI tract Induces glyconeogenesis/hyperglycemia regulates GI motility
G	gastrin	stimulates gastric release of HCl
PP	pancreatic polypeptide	inhibits release of HCl and stimulates release of enzymes by gastric chief cells

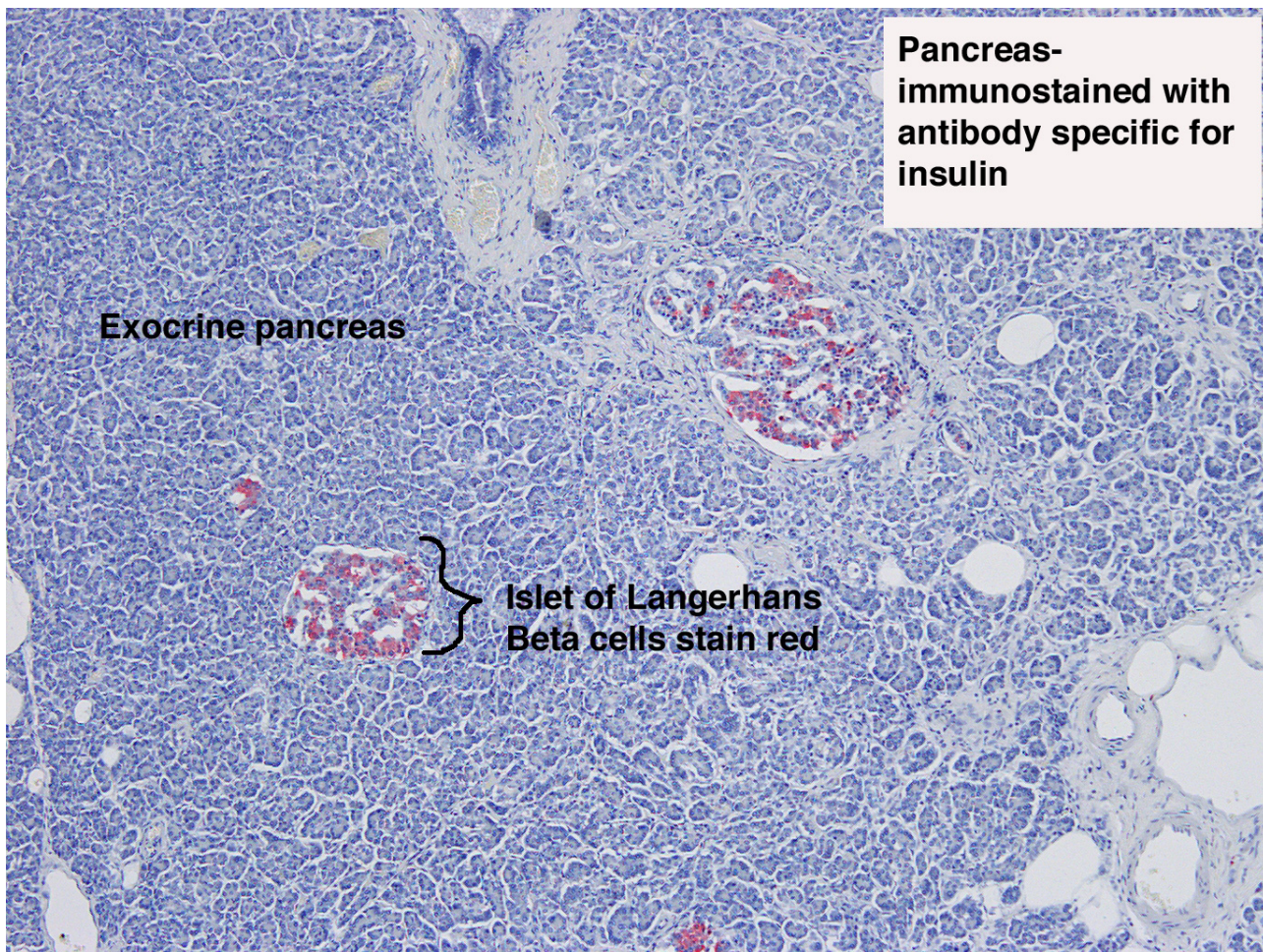


Image 11: 10X image of pancreas

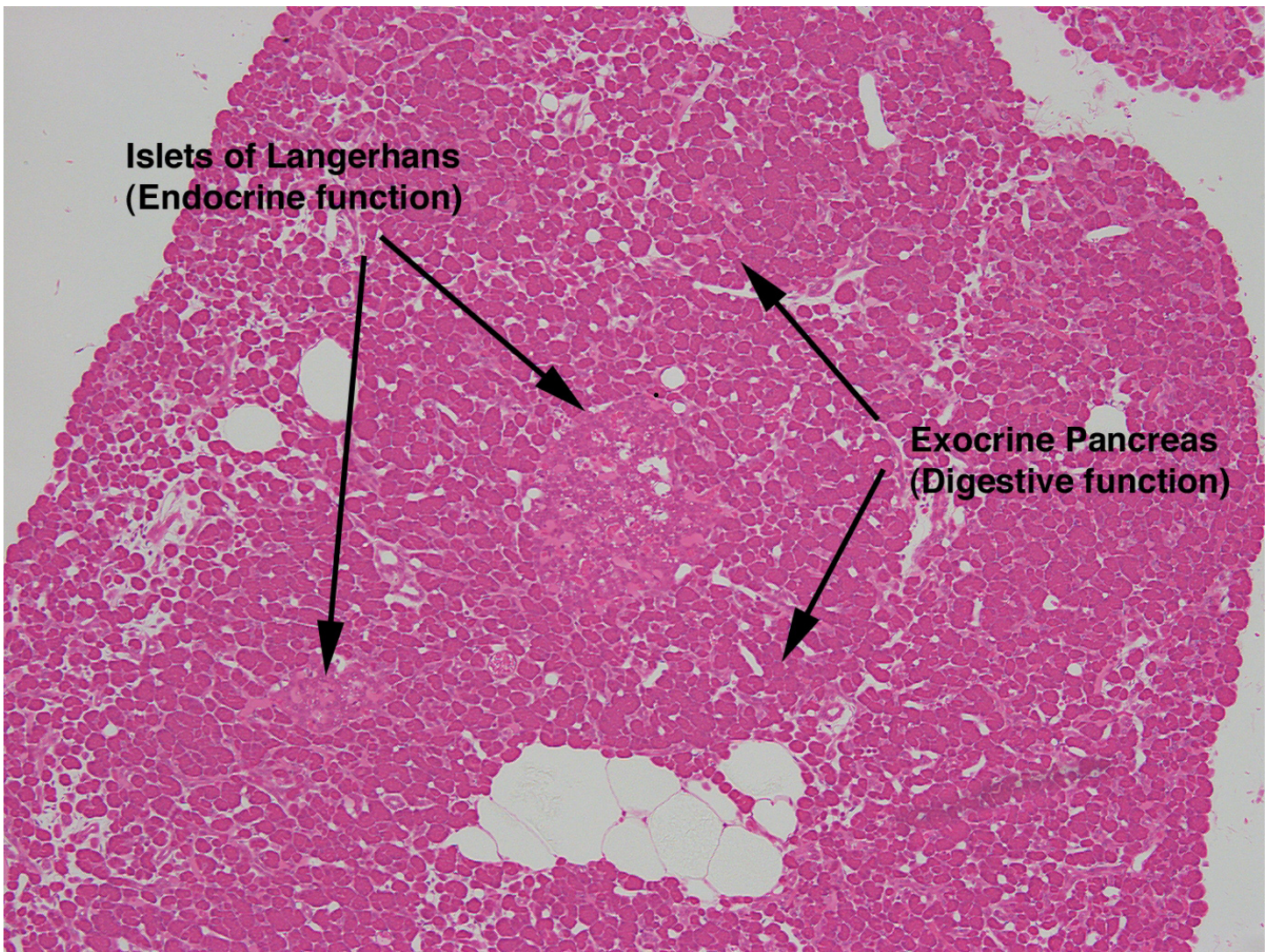


Image 12: 10X image of pancreas

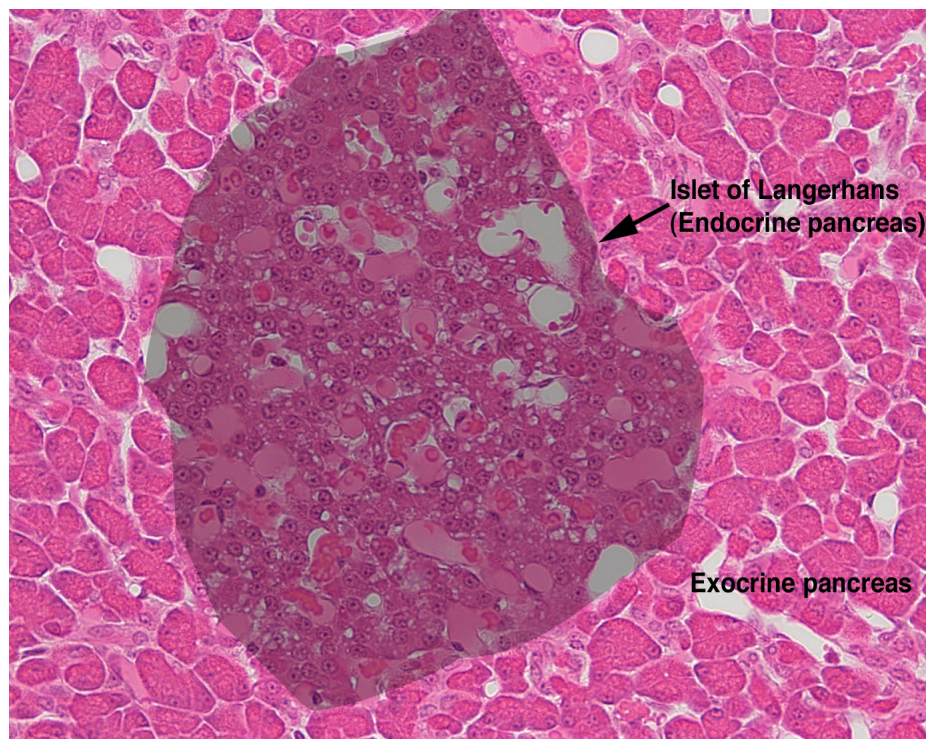


Image 13: 40X image of pancreas with endocrine pancreas area shaded

Some Clinical Correlations

Pituitary gland

Pituitary adenomas - tumors in the anterior pituitary - may suppress hormonal production

Diabetes insipidus - lesion of hypothalamus or posterior pituitary - reduced antidiuretic hormone (ADH), therefore decreased water absorption by kidneys, so patient drinks alot

Thyroid gland

Graves' disease - hyperplasia of follicular cells, hyperthyroidism, enlarged thyroid gland

Goiter - enlarged thyroid gland due to insufficient dietary intake of iodine

Hypothyroidism - slowed metabolism results in excess sleep, muscle sluggishness, slowed heart rate, constipation

If a thyroidectomy is to be performed, the surgeon must locate the external laryngeal and recurrent laryngeal nerves. If these are damaged, the patient may become dysarthric.

Parathyroid glands

Hyperparathyroidism - excess of PTH causes increased blood calcium levels, loss of bone minerals and possibel kidney stones

Hypoparathyroidism - decreased secretion of PTH, causes low blood calcium levels, muscle cramps, tetany, mental confusion

Suprarenal glands

Addison's disease - decreased production of adrenocortical hormones due to the destruction of the adrenal cortex. Steroid treatment is needed or death will occur.

Cushing's disease - excess production of ACTH causes excess production of cortisol resulting in obesity, especially in face, neck and trunk, and muscle wasting, impotency in males, amenorrhea in females.

Pancreas

Diabetes mellitus - hyperglycemic metabolic disorder results in either lack of production of insulin from the beta cells in the endocrine pancreas (type I diabetes), or defective insulin receptors on target cells (type II diabetes).